Layer 2 Network Security

DHCP Snooping

Concept Overview

DHCP Snooping acts as a Layer 2 firewall between untrusted hosts and DHCP servers. It validates DHCP messages received from untrusted interfaces and builds a dynamic binding database known as the **DHCP Snooping Binding Database**. This database tracks the client's MAC address, assigned IP address, lease time, binding type, VLAN number, and switch interface point of entry.

Switch ports are categorized into two logical operational states:

- **Trusted Ports:** Connect to legitimate DHCP servers or network infrastructure (switches/routers). DHCP responses (OFFER, ACK, NAK) are permitted to flow across these interfaces.
- **Untrusted Ports:** Connect to end-user devices, hosts, or untrusted network segments. These ports are strictly prohibited from sourcing DHCP server response traffic.

Mitigated Attacks

- **Rogue DHCP Server Attacks:** An unauthorized device responds to client DHCP requests first, handing out malicious IP settings (e.g., setting the attacker's IP as the default gateway or DNS server) to perform Man-in-the-Middle (MitM) traffic inspection.
- **DHCP Starvation Attacks:** An attacker floods the network with DHCP requests containing randomized MAC addresses, exhausting the DHCP server's IP pool and causing a denial-of-service (DoS) for legitimate endpoints.

Lab Configuration

Enable DHCP Snooping globally

Switch(config)# ip dhcp snooping

Specify target VLANs for DHCP inspection

```
Switch(config)# ip dhcp snooping vlan 10,20

Switch(config)# no ip dhcp snooping information option 82

Switch(config)# interface GigabitEthernet0/1
Switch(config-if)# description Uplink to Core / DHCP Server
Switch(config-if)# ip dhcp snooping trust
Switch(config-if)# exit

Switch(config)# interface range GigabitEthernet1/0/1 - 24
Switch(config-if-range)# description Access Ports
Switch(config-if-range)# ip dhcp snooping limit rate 15
Switch(config-if-range)# exit
```

Dynamic ARP Inspection (DAI)

Concept Overview

Dynamic ARP Inspection (DAI) security feature validates Address Resolution Protocol (ARP) packets within a network network infrastructure. DAI intercepts, logs, and discards ARP packets with invalid MAC-to-IP address bindings.

DAI operates by utilizing the **DHCP Snooping Binding Database** (or statically configured ARP Access Control Lists). When an ARP request or response arrives on an untrusted port, the switch compares the sender IP address and hardware MAC address against the binding database entry:

- If the IP-to-MAC mapping matches the database, the frame is forwarded.
- If the IP-to-MAC mapping does not match, the ARP packet is dropped, preventing cache poisoning.

Mitigated Attacks

- **ARP Poisoning / Spoofing:** Attackers send gratuitous or crafted ARP messages to map their hardware address to a legitimate IP address (such as the default gateway), enabling traffic redirection, passive sniffing, or active inline modification.
- **Man-in-the-Middle (MitM) Redirection:** Disrupts traffic flows by altering neighbor ARP caches across the broadcast domain.

Lab Configuration

DAI requires DHCP Snooping enabled to construct its binding database

Switch(config)# ip dhcp snooping

Switch(config)# ip dhcp snooping vlan 10

Enable DAI on specified VLANs

Switch(config)# ip arp inspection vlan 10

Configure optional validation checks (SIP/DIP/MAC verification)

Switch(config)# ip arp inspection validate src-mac dst-mac ip

Switch(config)# interface GigabitEthernet0/1

Switch(config-if)# description Trunk to Core Switch

Switch(config-if)# ip arp inspection trust

Switch(config-if)# exit

Switch(config)# interface range GigabitEthernet1/0/1 - 24

Switch(config-if-range)# ip arp inspection limit rate 15

Switch(config-if-range)# exit

Storm Control

Concept Overview

Storm Control measures incoming traffic levels on a switch port over a 1-second interval and compares the measured traffic against a user-defined threshold. It operates independently for broadcast, multicast, and unicast traffic streams.

When the configured threshold is reached, the switch takes preventative action: it drops incoming traffic of that specific type until traffic rates drop below a secondary lower threshold (or until the monitoring interval resets). This prevents excess broadcast or multicast frames from consuming hardware queues and link bandwidth across the entire Layer 2 domain.

Mitigated Attacks

- **Broadcast Storms:** Caused by hardware failures, physical switching loops, or misconfigured spanning-tree topologies.
- **Denial-of-Service (DoS) via Flooding:** Prevents compromised or malicious hosts from saturating port capability and CPU processing using high-volume broadcast, unknown multicast, or unknown unicast traffic bursts.

Lab Configuration

Switch(config)# interface range GigabitEthernet1/0/1 - 24

Switch(config-if-range)# storm-control broadcast level 20.00 10.00

Switch(config-if-range)# storm-control multicast level 15.00

Switch(config-if-range)# storm-control unicast level 10.00

Specify corrective action (drop traffic by default; optionally generate SNMP trap or err-disable)

Switch(config-if-range)# storm-control action trap

Switch(config-if-range)# exit

IPv6 Router Advertisement (RA) Guard

Concept Overview

IPv6 networks utilize Neighbor Discovery Protocol (NDP) and Router Advertisements (RAs) to auto-configure hosts (SLAAC) and advertise local default gateways. RA Guard acts as a Layer 2 inspection engine for IPv6 control frames.

RA Guard inspects incoming ICMPv6 packets on switch access ports and drops IPv6 Router Advertisement (Type 134) and Router Redirect (Type 137) messages sourced from unauthorized interfaces. Port roles are explicitly assigned as either **Client** (untrusted host ports) or **Router** (trusted interfaces connected to legitimate IPv6 routers).

Mitigated Attacks

- **Rogue IPv6 Router Advertisements:** Unauthorized routers or compromised hosts sending fake RA messages, causing clients to reconfigure default gateways and IPv6 prefixes automatically.
- **IPv6 MitM & DoS:** Redirecting external IPv6 traffic flows through an attacker's system or disabling host IPv6 connectivity by advertising dead-end routes.

Lab Configuration

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```
 Define RA Guard Policy for host access ports
Switch(config) ipv6 nd inspection policy Host-RA-Policy
Switch(config-nd-policy) device-role client
Switch(config-nd-policy) exit

 Define RA Guard Policy for trusted router ports
Switch(config) ipv6 nd inspection policy Router-RA-Policy
Switch(config-nd-policy) device-role router
Switch(config-nd-policy) exit

 Apply RA Guard policy to client access interfaces
Switch(config) interface range GigabitEthernet1/0/1 - 24
Switch(config-if-range) ipv6 nd raguard attach-policy Host-RA-Policy
Switch(config-if-range) exit

 Apply RA Guard policy to core router interface
Switch(config) interface GigabitEthernet0/1
Switch(config-if) ipv6 nd raguard attach-policy Router-RA-Policy
Switch(config-if) exit
```

Port Security

Concept Overview

Port Security limits access on an interface by dynamically or statically learning and restricting the specific Ethernet MAC addresses permitted to send frames into a switch port.

When port security is enabled, the switch maintains a record of authorized MAC addresses associated with that port. If an unapproved MAC attempts to enter the port, or if the number of learned addresses exceeds the maximum configured limit, a security violation is triggered.

Security violation modes include:

Protect: Drops packets with unknown source MACs until secured addresses are cleared. No log entries are generated.

Restrict: Drops non-compliant packets, increments the security violation counter, and generates an SNMP trap/syslog message.

Shutdown (Default): Immediately places the interface into the `err-disable` state, shutting down link operations until administrative intervention or automated recovery occurs.

Mitigated Attacks

MAC Address Flooding: Attacks attempting to fill the switch CAM (Content Addressable Memory) table with fake MAC addresses, forcing the switch into fail-open unicast flooding mode.

Unauthorized Device Attachment: Rogue physical devices plugging into active, unauthenticated physical switch ports.

Lab Configuration

Port Security requires statically configured Access or Trunk mode

Switch(config) interface range GigabitEthernet1/0/1 - 10

Switch(config-if-range) switchport mode access

Switch(config-if-range) switchport port-security

Define maximum allowed MAC addresses per port

Switch(config-if-range) switchport port-security maximum 2

Enable Sticky MAC Learning (dynamically learns and adds to running-config)

Switch(config-if-range) switchport port-security mac-address sticky

Define Violation Action (Protect | Restrict | Shutdown)

|---|Switch(config-if-range) switchport port-security violation restrict

Optional: Configure automatic recovery from err-disable state (if using Shutdown mode)

Switch(config) errdisable recovery cause psecure-violation

Switch(config) errdisable recovery interval 300

Summary Matrix of Controls

Security Feature	Operational Layer	Primary Mitigation Focus
DHCP Snooping	Layer 2 / Layer 7	Rogue DHCP Servers, IP/MAC Spoofing
Dynamic ARP Inspection	Layer 2	ARP Cache Poisoning, MitM Attacks
Storm Control	Layer 2 Physical	Broadcast Loops, Denial of Service
RA Guard	Layer 2 / ICMPv6	Rogue IPv6 Routers, SLAAC Hijacking
Port Security	Layer 2 Physical	CAM Table Flooding, Rogue Hardware